MLOps Coaching Program

Comprehensive Syllabus for Freshers

Building a Career in Machine Learning Operations
Updated: August 2026

Program Overview

This 16-week intensive coaching program is designed for freshers to master MLOps and launch a successful career in Machine Learning Operations. The curriculum bridges the gap between machine learning and software engineering, covering essential skills from fundamentals to production deployment.

Learning Outcomes

- Understand ML fundamentals and MLOps principles
- Master Python programming and version control
- Build containerized applications with Docker
- Implement CI/CD pipelines for ML projects
- Deploy and serve ML models in production
- Monitor, log, and maintain ML systems
- Implement ML infrastructure best practices

Prerequisites

Basic programming knowledge, understanding of command line, basic Python familiarity, basic git concepts, understanding of how ML models work

Course Structure & Duration

Duration: 16 weeks (20-25 hours/week)
Format: Live sessions + Self-paced modules + Hands-on projects

WEEK	MODULE	TOPICS	KEY PROJECTS
1-2	MLOps Fundamentals	What is MLOps, ML lifecycle, Model training vs deployment, Why MLOps matters	Environment setup

3-4	Python for MLOps	Python basics, NumPy & Pandas, Data manipulation, Virtual environments	Data cleaning project
5-6	Version Control & Git	Git fundamentals, GitHub workflow, Branching strategies, Pull requests	Git workflow project
7-8	Docker & Containerization	Docker basics, Dockerfile creation, Multi-stage builds, Docker compose	Containerize ML app
9-10	CI/CD Pipelines	GitHub Actions, Pipeline design, Testing automation, Code quality	Build ML CI/CD
11-12	Model Deployment	FastAPI, Model serialization, API versioning, Load balancing	Deploy model as API
13-14	Monitoring & Production	Logging, Monitoring, Performance tracking, Infrastructure as Code	Monitoring setup
15-16	Advanced & Capstone	Model registry, Feature stores, A/B testing, Security, Career prep	Capstone project

Assessment & Evaluation

- Weekly coding assignments (40%)
- Mini-projects per module (30%)
- Final capstone project (20%)
- Participation & code reviews (10%)

Tools & Technologies Covered

- Python, Git, Docker, GitHub Actions, FastAPI, PostgreSQL
- Cloud platforms: AWS (EC2, S3, Lambda), Google Cloud (optional)
- Monitoring: Prometheus, Grafana, ELK Stack
- ML frameworks: Scikit-learn, TensorFlow, PyTorch

Career Outcomes

Upon completion, graduates can pursue roles such as:
- MLOps Engineer
- Machine Learning Engineer
- Data Engineer
- DevOps Engineer (ML focus)
- ML Platform Engineer

Learning Resources

- Official documentation and tutorials
- Real-world projects & open-source
- Industry case studies
- Weekly live Q&A; sessions
- Mentorship from professionals